LESSON PLAN

ENGINEERING SYSTEM (21)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	2	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to liquid rocket engines its advantages and disadvantages
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to liquid rocket engines its advantages and disadvantages. Introduction
5. Specific Objectives	21.1.1.1 Trainee will be able to understand Rockets, types of its functional details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of PPT slides on Liquid Rocket Engine components; Safety: Maintain distance from projector; Supervision: Instructor monitors trainee engagement.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a rocket engine?
(b) Name two types of rocket engines based on propellant state.
(c) Why are liquid rocket engines preferred for upper stages?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Liquid Rocket Engine • Propellant Storage Systems • Fuel and Oxidizer Mixing • Combustion Chamber Function • Thrust Vector Control	L & Ds	PPT, IB	15 min
(b)	• Advantages of Liquid Propulsion • Disadvantages of Liquid Propulsion • Comparison with Solid Rockets • Throttleability Features • Complexity and Maintenance Issues	L & Ds	PPT, PDF	20 min
(c)	• Fuel Types Used in Liquid Engines • Oxidizer Types Used in Liquid Engines • Engine Start and Shutdown Procedures • Specific Impulse Concepts • Application in Launch Vehicles	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) List three advantages of liquid rocket engines over solid rocket motors.
(b) Explain why liquid engines are more complex to operate than solid engines.
(c) Describe the role of the turbopump in a liquid rocket engine.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operational flexibility of liquid rocket engines with solid rocket motors for future mission planning.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min